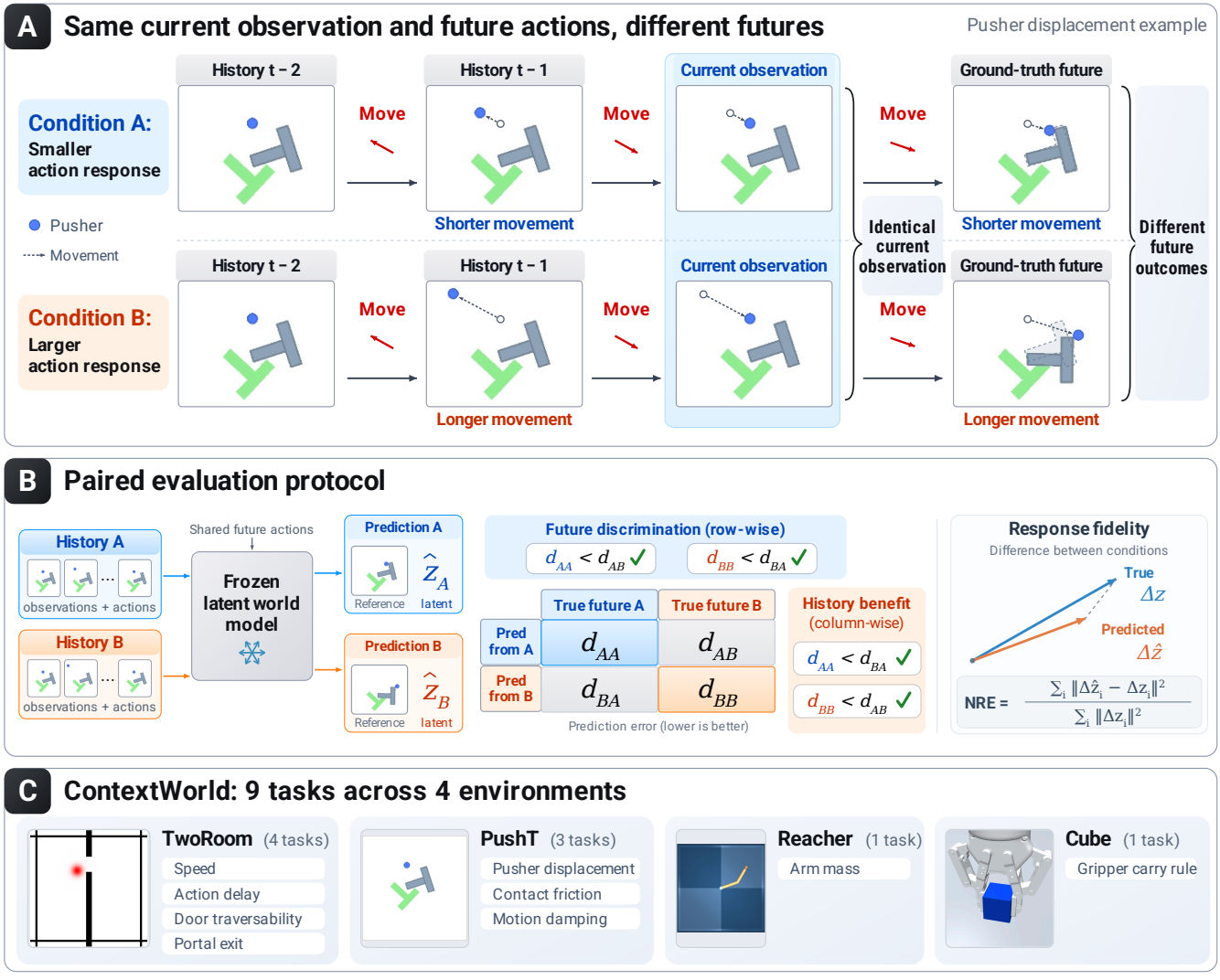

A
Same current observation and future actions, different futures
Pusher displacement example
Condition A:
Smaller action response
Pusher
Movement
Condition B:
Larger action response
History t − 2
History t − 1
Current observation
Ground-truth future
Move
Shorter movement
Longer movement
Identical current observation
Different future outcomes
B
Paired evaluation protocol
History A
History B
observations + actions
Shared future actions
Frozen latent world model
Prediction A
Prediction B
Reference
latent
Future discrimination (row-wise)
True future A
True future B
Pred from A
Pred from B
Prediction error (lower is better)
History benefit
(column-wise)
Response fidelity
Difference between conditions
True
Predicted
NRE =
C
ContextWorld: 9 tasks across 4 environments
TwoRoom (4 tasks)
Speed
Action delay
Door traversability
Portal exit
PushT (3 tasks)
Pusher displacement
Contact friction
Motion damping
Reacher (1 task)
Arm mass
Cube (1 task)
Gripper carry rule